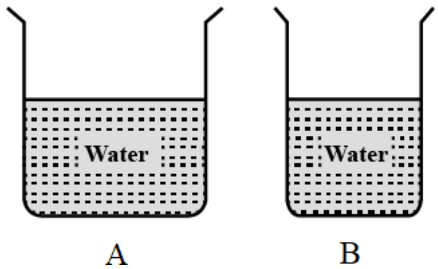


FREEDOM INTERNATIONAL SCHOOL
WORKSHEET(SOLUTIONS)
PHYSICS
MECHANICAL PROPERTIES OF FLUIDS

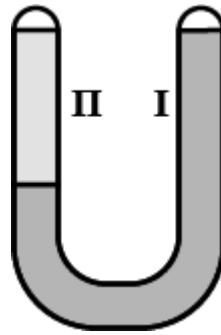
CLASS: XI

MAX MARKS: 10

TIME: 15 MINS.

1.	From the following figures, the correct observation is  A B (a) the pressure at the bottom of tank A is greater than that at the bottom of B (b) the pressure at the bottom of the tank A is smaller than that at the bottom of B (c) the pressure depends on the shape of the container (d) the pressure at the bottoms of A and B is the same Ans: (d) the pressure at the bottoms of A and B is the same.	1
2.	The radius of one arm of a hydraulic lift is four times of the radius of another arm. What force should be applied on the narrow arm to lift 100 kg? (a) 26.5 N (b) 62.5 N (c) 6.25 N (d) 8.3 N Ans: (b) 62.5 N By Pascal's law, $\frac{F}{A} = \frac{f}{a}$ $\text{or } f = \frac{Fa}{A} = \frac{100g \times (\pi r)^2}{(\pi \times 4r)^2}$ $= 6.25g = 62.5N$	1
3.	The ratio of the terminal velocities of two drops of radii R and R/2 is (a) 2 (b) 1 (c) 1/2 (d) 4 Ans: (d) 4 Terminal velocity, $v_T \propto r^2$ $\therefore \frac{(v_T)_1}{(v_T)_2} = \left(\frac{R}{R/2} \right)^2 = 4$	1

4. A U-tube of uniform cross-section is partially filled with a liquid I. Another liquid II which does not mix with liquid I is poured into one side. It is found that the liquid levels of the two sides of the tube are the same, while the level of liquid I has risen by 2 cm. If the specific gravity of liquid I is 1.1, the specific gravity of liquid II must be



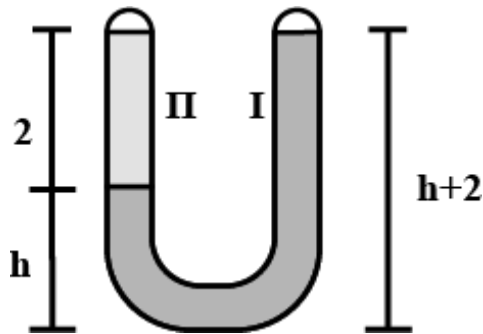
(a) 1.12

(b) 1.1

(c) 1.05

(d) 1.0

Ans: (b) 1.1



It is clear that the height of liquid column I in right limb is $(h + 2)$ cm and the height of liquid column II in left limb is 2.0 cm and of liquid column I is h cm .

$$\therefore 2 \times \rho_2 \times g + h \times 1.1 \times g = (2 + h) \times 1.1 \times g$$

$$\Rightarrow 2\rho_2 = 2.2$$

$$\Rightarrow \rho_2 = 1.1$$

5.	A mercury drop of radius 1 cm is broken into 10^6 droplets of equal size. The work done is ($\rho = 35 \times 10^{-2} \text{ N/m}$) (a) $4.35 \times 10^{-2} \text{ J}$ (b) $4.35 \times 10^{-3} \text{ J}$ (c) $4.35 \times 10^{-6} \text{ J}$ (d) $4.35 \times 10^{-8} \text{ J}$ Ans: (a) $4.35 \times 10^{-2} \text{ J}$ Initial radius of drop, $R = 1 \text{ cm} = 10^{-2} \text{ m}$ suppose, r is the radius of droplets since, volume will be remain constant. $\frac{4}{3}\pi R^3 = 10^6 \times \frac{4}{3}\pi r^3$ [since, number of droplets = 10^6] $r = \frac{1 \times 10^{-2}}{10^2}$ Increase in surface area, $\Delta S = 10^6 \times 4\pi r^2 - 4\pi R^2$ $= 10^6 \times 4 \times \pi \times 10^{-8} - 4\pi \times 10^{-4}$ $= 4 \times 9.9 \times \pi \times 10^{-3}$ Work done = increase in surface area $\times$ Surface tension. $= 4 \times 9.9 \times \pi \times 10^{-3} \times 35 \times 10^{-2}$ $= 4.35 \times 10^{-2} \text{ J}$	1
6.	Dynamic lift is related to (a) Bernoulli's theorem (b) Archimedes' principle (c) Equation of continuity (d) Pascal's law Ans: (a) Bernoulli's theorem	1
7.	The surface tension of a soap solution is $25 \times 10^{-3} \text{ N/m}$. The excess pressure inside a soap bubble of diameter 1 cm is (a) 10 Pa (b) 20 Pa (c) 5 Pa (d) none of these Ans: (b) 20 Pa The excess pressure is calculated as $\Delta P = \frac{4T}{r}$ $\Delta P = \frac{4 \times 25 \times 10^{-3}}{0.5 \times 10^{-2}} = 20 \text{ N/m}^2$	1
8.	If the liquid neither rises nor falls in a capillary tube, then angle of contact is (a) 0° (b) 180° (c) 90° (d) 45° Ans: (c) 90°	1

